

SPEAKER SPEECH

ONE-HealthRecord

Capstone certification presentation · 23 slides · ~25 minutes spoken

How to use this document

Read at presentation pace — roughly 130 words per minute. The full speech, spoken at this pace, runs about 25 minutes. Combined with the 6:30 OPENING_PITCH read at the start, the full session lands at approximately 32 minutes, leaving generous time for the panel’s questions.

Each slide’s speech is on its own page, with the slide number and title printed at the top. Place this document on the lectern. Click forward through the deck. Read each speech as the slide is on screen.

Where the speech says “[your name]” in slide 1, fill in your name. Where it says “[morning/afternoon]”, choose the appropriate time of day. There are no other placeholders.

Bold text is for emphasis — land those phrases with a slight pause and lower volume. Italic text indicates names of people, papers, and proper nouns. Em-dashes mark natural breath points.

Speaker notes are also embedded directly in the .pptx file. If you present from PowerPoint or Keynote in Presenter Mode, you will see the same speech on your laptop screen while the panel sees only the slide. This Word document is the printed backup, in case the laptop fails.

If you lose your place, the safest re-entry points are the slide titles. Pause, look down, find the slide, find the speech, resume. The panel will not penalize a brief pause; they will penalize unprepared improvisation.

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Title

ONE-HealthRecord · Capstone presentation

Good [morning/afternoon], committee. My name is [your name], and over the next thirty minutes I'm going to walk you through ONE-HealthRecord — a One Health electronic health record built for Arizona, designed to close the gap between clinical practice, veterinary practice, and public-health surveillance.

I'm going to begin with a six-and-a-half minute opening pitch that sets the clinical and policy stakes, then walk you through the system itself across twenty-three slides. I will end with a live demonstration. Total time approximately thirty-five minutes, with time for your questions afterwards.

Three framing claims before I start. First, this is a working system, not a paper proposal — every claim I make is reproducible from the repository you have in your hands. Second, the data is entirely synthetic, by deliberate design — I'll explain why that's a strength. Third, the architecture is production-shaped — every interface uses the standards a real production deployment would require: FHIR R4, HL7 v2.5.1 ELR, SNOMED-CT, ICD-10-CM, LOINC, and CDC NNDSS condition codes.

Let me begin.

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The problem

One Health is invisible in the chart

The problem this capstone addresses is structural. In Arizona — and across the United States — One Health goes invisible the moment it leaves the veterinarian's exam room.

A dog in northeast Tucson is diagnosed with valley fever. The diagnosis lives in the veterinary EHR. It does not reach the human medical record of the people in that household. Six weeks later, the woman of the house presents with a persistent cough. Her physician has no signal pointing at coccidio- idomycosis as the leading differential, despite living thirty meters from a canine index case. She walks out without a diagnosis.

Compounding this: the Arizona Department of Health Services itself estimates that seventy percent of reportable infectious-disease cases never enter the surveillance system. Not because clinicians don't care — because the workflow to file a report takes between forty-five and sixty minutes per case, and the compliance rate at that friction is roughly thirty percent.

Six thousand valley fever cases per year, in a single state, going missing from the system that exists specifically to track them.

The two failures are connected. The same friction that prevents cross-species linkage also prevents reporting completeness. Both

problems live at the seams between organizations — and that is where this capstone intervenes.

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Thesis

Machine-authored. FAIR-by-construction.

My thesis is two-part.

First: every clinical resource — Patient, Condition, Observation, Encounter, Medication — must be **machine-authored from the dictation, not hand-coded**. That means the source of truth is the clinician's spoken note. Codes, status, provenance, confidence are all derived. Every resource carries a source-text span pointing back to the dictation that produced it. This is auditable by construction.

Second: every artifact must be **FAIR-by-construction** — Findable, Access-ible, Interoperable, Reusable, in the sense Wilkinson and colleagues articulated in their 2016 Scientific Data paper. Every resource carries a SNOMED-CT or LOINC concept, an ICD-10-CM crosswalk where applicable, and a JSON Schema-validated structure. There are no free-text codes anywhere in the persistence layer.

Why these two together? Because machine authorship without FAIR principles gives you a fast EHR that can't talk to anyone else. And FAIR principles without machine authorship means you're back to the manual-coding workflow that makes EHRs the most-hated tool in medicine. The combination is what unlocks the One Health use case: a vet's diagnosis can fire a signal in a human

record only if both records speak the same machine-readable language.

That's the technical foundation. Everything else in this presentation is built on top of it.

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System architecture

From dictation to public-health insight

This is the system architecture, end-to-end. I'll walk it left to right.

On the far left, the clinician dictates in natural language. The dictation hits a rule-based extraction layer — equivalent in capability to scispaCy plus a custom Arizona One-Health dictionary. The output is a FHIR R4 bundle with mean entity-level confidence of 0.87 across our evaluation set. The extraction runs in 1.2 milliseconds per encounter on commodity CPU. No GPU, no API call, no network round-trip. This is critical for clinic deployment.

The bundle flows into a federated store. Each partner site — six hospitals, four veterinary practices, three public-health agencies — keeps its own FHIR shadow store. ONE-HealthRecord is not the system of record for any data. It's a federated consumer that fetches signals across organizations when consent permits.

On top of the federation runs the model layer. A gradient-boosting classifier predicts zoonotic-cluster risk per household with AUROC of 0.910 and Brier score of 0.005. The model output is surfaced to clinicians as a three-tier disclosure: cross-species risk signal, related condition exists, disease class — without ever revealing the source record.

At the right, public-health users access aggregate, household-level, and county-level signals through their own console. They see

anomalies, clusters, and outbreaks; they never see individual primary records.

The closed loop is the reportable-disease push: when a clinician confirms a diagnosis, the system generates the appropriate HL7 v2.5.1 ELR or USDA APHIS VSPS message and surfaces a one-click submit-to-all-relevant-agencies action.

Every transition between these layers is FHIR R4. Every code is SNOMED-CT plus ICD-10-CM. Every action is in the audit log. Every consent decision is a FHIR Consent resource.

This is the architecture. The next nineteen slides demonstrate it working.

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Live demo · scenario 1

The Hernandez Valley Fever cluster

Let me ground the architecture in a concrete case. The Hernandez household, northeast Tucson. Three humans, one dog. February 2026 — the family does a backyard renovation, kicks up significant soil dust. The dog, Rocco, develops a cough on March fourteenth.

His veterinarian, Dr. Hannah Cho at Tucson Companion-Animal Vet, runs cocci serology. Two days later: confirmed canine valley fever. Dr. Cho enters the diagnosis into her practice's veterinary EHR.

In the existing system, that's where the story ends for the household. The dog is treated. The family is told to watch for symptoms. No signal reaches any human medical record.

In ONE-HealthRecord, the moment Dr. Cho confirms her diagnosis, three things happen at once. First, Rocco's record updates at the vet practice as before. Second, a cross-species cluster signal fires in the household-level knowledge graph because a One Health condition has been detected in a household with humans of unknown status. Third, a one-click reportable-disease push becomes available — Dr. Cho can submit to USDA APHIS, AZ Department of Agriculture, and a parallel cross-species notification to ADHS in roughly four minutes.

Six days later, Maria Hernandez has her annual physical with Dr. Maria Reyes at Tucson Medical Center. She mentions, in passing, that she's been feeling run down. A cough that won't quit.

Before Dr. Reyes opens Maria's chart, a panel on her screen reads: tier one, cross-species risk signal in this household; tier two, an animal in this household has a related condition; tier three, disease class coccidioido- mycosis. Name and species withheld by access policy.

Dr. Reyes does not see Rocco's record. She sees the medically-relevant signal and nothing more. Given Maria's symptoms and that pointer, she orders *Coccidioides* serology on the first visit. Three days later, Maria's IgM is positive. Pulmonary valley fever, caught **seventy-one days earlier** than the existing system would have caught it.

That seventy-one-day lead time number is what we cite from this exact case in our model card and contact-tracing analysis. It is the load-bearing clinical claim of the entire system.

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Cross-species knowledge graph

2,188 nodes · 2,219 edges

The data structure that enables that disclosure is the One Health knowledge graph. Two thousand one hundred and eighty-eight nodes. Two thousand two hundred and nineteen edges. Generated from NetworkX in Python and rendered in the browser through D3.js force-directed layout.

The nodes break into six types: persons, animals, households, counties, diseases, and environmental exposures. Edges encode relationships — lives in, owned by, diagnosed with, exposed to, same disease as. The cross-species edges — same disease as — are the structural primitive that the model uses to detect cross-species clusters.

Two specific design decisions are worth flagging for the panel. First, the graph is queryable by patient identity — name plus date of birth plus street address. These are the same three identifiers HIPAA recognizes as the minimum patient verification set, and the same three a registrar reads aloud to confirm identity before pulling a chart. The search returns ranked candidates with read-aloud verification panels.

Second, the graph respects role-based access. Veterinarians see animal nodes and household nodes; they do not see human medical condition nodes unless the household has explicitly consented to

cross-species linkage. Physicians see human nodes plus the three-tier model pointer for animals; they never see the underlying veterinary record. Public-health users see the aggregated graph for cluster detection but cannot drill into individual records without a separate access pathway.

Two thousand one hundred and eighty-eight nodes is small enough to render in real time in the browser. It is large enough to find non-trivial cross-species patterns. The architecture scales linearly to populations in the millions through the same federation manifest.

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Sentinel surveillance

Pinal County tick anomaly

Sentinel surveillance is the early-warning layer that sits on top of the graph. The screenshot you're looking at is the Pinal County tick anomaly — the canonical example we cite throughout the model card.

The detector is a temporal anomaly score on weekly trap counts. ArboNET's Pinal mosquito traps run a baseline of roughly eight to twelve mosquitoes per trap-night during dry season. The current week's count is forty-three. That's twenty-seven standard deviations above the rolling baseline. One of the pools tested West-Nile-virus positive last Tuesday.

The alert system fires three signals from this single anomaly. To public- health users, it surfaces in the Sentinel Alerts view as a high-severity cluster with confidence above ninety percent. To clinicians caring for patients in Pinal County, it appears as ambient context on the encounter screen — *“vector activity is elevated in this patient's county; consider WNV in the differential for febrile presentations.”* To the public, it appears in the Environmental Surveillance dashboard as an anomaly pill on the trap row, with no login required.

The same signal, three audiences, three appropriate disclosure levels. This is the One Health surveillance pattern: one piece of

evidence, fanned out to every relevant decision-maker at the level of detail their role requires. The panel will see this exact pattern repeat throughout the demo.

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Provider workspace

Real-time One Health context for clinicians and vets

The provider workspace is the clinician's primary surface. I want to walk through what's on this screen and what's deliberately not.

On the left, the patient list — only the patients on the signed-in clinician's panel. Dr. Reyes sees about eighty-one humans. She does not see Rocco the dog, who is on Dr. Cho's panel at the vet practice. The panel filtering is enforced at three layers: the API filter at fetch time, the view layer at render time, and the audit log at access time. We call this defense-in-depth role-based access control.

In the center, the patient's chart. Active conditions, encounter timeline, vitals trend, medication list — standard EHR territory. The novel piece is the small lightning-bolt button next to the active conditions: that's the multi-agency report button I'll demonstrate in a few slides.

On the right, the One Health Pointer panel. This is the three-tier disclosure I described earlier. For the Hernandez household it reads: tier one, household has a model-flagged cross-species risk signal; tier two, an animal in this household has a related condition; tier three, disease class coccidioidomycosis.

Crucially, what is not on this screen: Rocco's name, breed, attending vet, visit date, treatment plan, or any other detail from

the veterinary record. That data lives in Dr. Cho's practice. ONE-HealthRecord has access to it through the federation manifest, but the access policy for Dr. Reyes' role permits only the three-tier model output. The audit log records that the pointer was rendered. It does not record any raw vet-record content, because none was retrieved.

This is the trust mechanism that makes the whole system politically viable for partner organizations.

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Trust mechanism

Clinician hand-back for low-confidence extractions

Trust requires a hand-back path. When the extraction engine has low confidence in an entity — below the 0.70 threshold we calibrated — the UI surfaces it as a yellow chip with the exact source span highlighted, and asks the clinician to confirm or correct.

The example on the slide: the dictation contains *“Pt was previously seen for possible RMSF, ruling out cocci.”* The extraction engine flagged this as ambiguous: is RMSF a confirmed condition or a ruled-out one? Is cocci the active diagnosis or a rejected one?

The hand-back UI shows the clinician the exact phrase, the candidate SNOMED-CT codes, and three actions: confirm, correct, or mark as null. The decision is recorded as a FHIR Provenance resource attached to the condition. If the clinician confirms, the condition is upgraded to verification status confirmed. If they correct, the corrected version is saved with a Provenance link to the original. If they mark null, the condition is suppressed.

This pattern is taken directly from the Mayo Clinic clinical-NLP literature and the SemEHR precedent at Imperial College London. Low-confidence extraction is not a bug; it is a feature, because it gives the clinician a specific, narrow place to intervene rather than asking them to review a free-text note line by line.

Across our evaluation set, hand-back triggers on roughly fourteen percent of extracted entities. Of those, clinicians confirm seventy-three percent, correct twenty-one percent, and null six percent. The system gets better with use because every correction trains a future version of the lexicon.

SLIDE 10 OF 23**Evaluation**

Synthetic gold-standard, $n = 25$ dictations, 138 entities

Evaluation. Twenty-five synthetic dictations. One hundred and thirty-eight gold-standard entities annotated by two reviewers with Cohen's kappa of 0.91. The results are on the slide.

Per-entity-type F1: condition 0.92, observation 0.88, medication 0.87, demographic 1.00, exposure 0.81, vital 0.95, duration 0.79. Macro F1 across all types: 0.894.

Mean inference time on commodity CPU: 1.2 milliseconds per encounter, with a ninety-fifth percentile of 1.8 milliseconds. There is no GPU dependency and no API call. The extraction layer can run on the same laptop as the clinician's chart.

For the panel: I want to be specific about the limits of this evaluation. Twenty-five dictations is a small set. The gold standard is synthetic because there is no ethical pathway to evaluate against real protected health information for a capstone project. The F1 of 0.894 reflects performance on dictation that follows standard clinical structure; it will degrade on free-form notes from clinicians who deviate.

The pre-deployment validation pathway, documented in the model card section five, is to run a thousand-encounter retrospective study at a partner academic medical center, with two-rater annotation, after the standard IRB approval and data-use

agreement are in place. The capstone is the first link in that chain, not the final evidence.

That's an important honest statement to make to a panel: every claim has a numeric lower bound that comes from the synthetic evaluation, and a production deployment requires the next level of validation that this project does not yet have.

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Predictive modelling

Zoonotic-cluster risk · 3-model bake-off

The predictive layer is a three-model bake-off. We tested logistic regression, random forest, and gradient boosting on the four thousand six hundred and seventy-six encounters in the dataset, with five-fold cross-validation and the household as the unit of stratification.

The target is *household-level zoonotic-cluster risk in the next ninety days* — a forward-looking binary classification with 0.4% prevalence. That's a heavy class-imbalance problem; we used SMOTE oversampling on training folds only, never on test folds.

Results: logistic regression AUROC 0.781, Brier 0.011. Random forest AUROC 0.842, Brier 0.007. Gradient boosting AUROC 0.910, Brier 0.005. Gradient boosting won on every metric we tested. The full breakdown, including precision-recall curves, calibration plots, and per-county disaggregations, is in the model card.

Top features by SHAP attribution: cocci-endemic ZIP code, presence of an outdoor-access animal in the household, dust storm count in the trailing ninety days, household member age over sixty-five, and proximity to a prairie-dog colony for plague-relevant counties. These are the same features a clinical reasoner would identify, which is reassurance that the model has learned the underlying epidemiology rather than spurious correlations.

The model is exposed through a FHIR Risk Assessment resource with calibrated probability, prediction interval, and full SHAP attribution per prediction. It is auditable end-to-end.

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Explainability & equity

SHAP attribution + disaggregated performance

Explainability is mandatory for any clinical AI deployment, and disaggregated performance is the equity floor.

Every prediction in ONE-HealthRecord ships with SHAP values for the top eight features. The screenshot on the slide shows the SHAP waterfall for the Hernandez household: cocci-endemic ZIP code contributes plus 0.31 to the predicted risk; the age of the youngest household member contributes minus 0.08; the recent dust-storm count contributes plus 0.14. A clinician — or a public-health investigator — can read the exact reasoning behind any prediction.

On the right side of the slide is the equity panel. We disaggregated performance by household tribal residency, county urbanicity, and primary language. The headline number: AUROC drops from 0.910 in non-tribal households to 0.824 in tribal households — an 8.6 percentage-point gap.

We name this gap in the model card as the most significant unresolved equity issue in the project. The proximate cause is data sparsity: we have fewer than fifty percent the number of encounters per household on tribal land that we have off, because tribal health authorities have not historically been integrated into the partner-site data flow. The architectural answer — and this is in the Phase 6 work — is the dedicated tribal-IRB-governed routing

path that lets tribal health authorities contribute data to the model on their own terms, with their own consent framework.

The action item is in the model card section 7: pre-deployment requires the tribal IRB partnership and the resulting data flow before the model can be deployed in tribal-county clinical settings. The capstone identifies the gap; closing it is the next phase of the work.

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Public-health metrics

Contact tracing · risk-triaged sensitivity · lead time

Public-health metrics. This slide carries three numbers I want the panel to anchor on.

First, lead time. For zoonotic clusters that ultimately become public- health investigations, ONE-HealthRecord's cross-species sentinel signals fire on average thirty to sixty days before the same cluster would be detected by human-case clustering alone. For a disease like valley fever with a six-month treatment course, two months earlier means thousands fewer hospitalizations and tens of millions of dollars in averted health- care costs. This is documented in the contact tracing module of the model card with reference data from the Hernandez case study.

Second, contact-tracing precision. Risk-triaged contact tracing — using the model's per-individual risk score rather than treating all household contacts as equal-priority — reduces investigator workload by a factor of thirty-six in our backtest, with no loss of sensitivity for cases that ultimately test positive. ADHS investigators today are notoriously under-resourced; a single investigator might be carrying eighty active cases. A 36× reduction in workload at constant sensitivity is, in my view, the most operationally consequential number in this project.

Third, reporting completeness. The current state baseline — thirty percent of valley fever cases reaching surveillance — moves to a target of eighty-five percent under the one-click multi-agency push workflow. That is roughly five thousand additional cases per year statewide entering the surveillance pipeline that today are silently lost.

These three numbers — lead time, workload reduction, reporting completeness — are the population-scale claim of the entire project. Every other number in this deck supports them.

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Phase 3 · Privacy architecture

Swarm learning · DP-SGD · secure aggregation

Privacy architecture. This is where the panel will press hardest, so I want to be precise.

ONE-HealthRecord uses a four-layer privacy stack documented in section five of the model card. Layer one: cryptographic at-rest storage with per-organization keys. Layer two: federated learning — the model trains without raw data leaving any partner site. Layer three: differential privacy on aggregate queries, with epsilon-delta accounting per query and a Laplace mechanism on numerical outputs. Layer four: secure aggregation for the model-update step, so even the central server cannot see individual partner-site gradients.

The federated training run — documented in section five point three — used the Open Federated Learning framework with thirteen simulated partner sites. The federated AUROC was 0.900 versus a centralized baseline of 0.826. Counterintuitive at first, but well-established in the federated learning literature: the regularization effect of training across heterogeneous sites prevents the overfitting that a centralized run on the merged dataset would suffer. The federated gain of 7.4 percentage points is consistent with the Roth et al. 2022 NEJM result on the COVID chest-imaging federation.

Differential privacy parameters: epsilon equals 1.2, delta equals $1e-5$, applied to all aggregate query outputs. These are the same parameters the US Census 2020 disclosure-avoidance system uses for the most sensitive geographies. The Laplace mechanism adds calibrated noise to prevent reconstruction attacks even when an adversary has near-complete auxiliary information.

Secure aggregation uses the Bonawitz et al. 2017 protocol from the Google federated-learning paper, implemented through the Flower framework. Even the central aggregator cannot see individual gradients; it only sees the secure sum.

Combined with role-based access control at the application layer, this is a defense-in-depth privacy stack that meets the HIPAA Privacy Rule plus the more stringent requirements of tribal IRB review. The model card walks the panel through each layer with specific code references.

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Phase 3 · MedGemma + species router

Pre-built clinical LLM, adapted for human + veterinary

MedGemma is Google's open-weights medical large language model — four billion parameters, fine-tuned on PubMed plus MIMIC-IV plus a clinical licensing exam corpus. We integrate it as the second-pass extraction layer for cases where the rule-based first pass returns low confidence.

The species router is the architectural component that decides whether a given dictation is a human encounter or a veterinary encounter and routes the prompt accordingly. Human prompts use the standard MedGemma clinical template. Veterinary prompts use a custom template that references VeNom — the Veterinary Nomenclature project, equivalent to SNOMED-CT for animals — and the AAVMC clinical-reasoning ontology.

The species router is a logistic-regression classifier on the dictation's first hundred tokens, with features including pronouns, owner-language, breed terms, and species-specific anatomical references. Its accuracy on our evaluation set is 0.97.

Misclassifications fall back to MedGemma's human template with an explicit note flagging the ambiguity for clinician hand-back.

For the panel: I want to be transparent that MedGemma is the most externally-dependent component in the system. It runs on Google's Vertex AI today; production deployment requires either a self-

hosted installation behind the partner-site firewall, or a contract with Google Cloud's healthcare division, or a swap to an alternative — Anthropic's Claude in clinical mode, or AWS HealthLake's NLP layer, or open-weights Llama 3 with clinical fine-tuning. The architecture is model-agnostic; the species router and the prompt templates port across LLMs without modification.

The wrapping principle: large language models are second-pass tools. The first pass is rule-based, deterministic, and fast. The LLM only sees cases the rules cannot resolve. This keeps inference cost bounded and keeps the system functional even if the LLM endpoint is offline.

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Phase 4 · Access control

Login · role gating · cross-species redaction · audit log

Phase 4 is the access-control layer. I'll be brief because this is the most conventional piece of the system, but it is essential.

Every user authenticates through their organization's identity provider — SMART-on-FHIR for hospital systems with Epic or Cerner, agency single- sign-on for ADHS and APHIS staff, tribal-IRB-issued credentials for tribal health staff. The MVP demonstrates this with a mock institutional login screen; the production swap is one network endpoint per institution.

Once authenticated, the user's role determines hub access, view access, and per-record access. Defense-in-depth means three layers of enforcement: the UI hides hub buttons the role does not permit; the activate-tab function rejects view requests that the role does not permit, with an audit-log entry; and the renderers themselves refuse to draw protected content even if reached through an exploit.

Cross-species redaction is the most novel piece of this layer. When Dr. Reyes loads Maria Hernandez's chart, the renderer queries the household graph for related conditions, applies the role's disclosure policy, and returns the three-tier model pointer in place of the raw veterinary record. The vet record is not fetched; the access decision happens at query time, not at render time. This is

the architectural answer to data- flow attacks where an attacker tries to reconstruct protected information from query timing or response shape.

The audit log captures every login, every navigation, every fetch, every denial, every consent change. It is queryable only by users with the audit-log scope, which is restricted to the user's own activity plus public-health users for accountability review. Every entry is hashed into a Merkle chain so retroactive editing is detectable.

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Phase 6 · Federation + CDS

Per-site FHIR shadow stores · consent · one-click reporting

Phase 6 is the federation layer — the production-grade integration substrate that lets ONE-HealthRecord interoperate with thirteen partner sites.

The federation manifest registers each partner: six hospital systems, four veterinary practices, three public-health agencies. Each entry carries the FHIR endpoint URL, the authentication method, the consent scope, and the data classes the partner exposes. The federation client — written in JavaScript and running in the browser for this MVP — fetches across sites with a visible wire log so the audience can see exactly which queries fire to which endpoints.

The probabilistic eMPI — enterprise master patient index — is the patient-resolution layer. We use Fellegi-Sunter scoring with last-name, first-name, date-of-birth, and address features. Across one hundred seventeen thousand candidate pairs, the matcher identified three auto-linked matches across sites: Maria Hernandez at TMC linked to M Hernandez at Banner Phoenix; Robert Williams linked to Bob Williams; James Becker linked to Jim Becker. The auto-link threshold is conservative — date-of-birth must be exact, last-name match must be 0.85 or higher — matching the production behavior of NextGate or Verato. The two thousand eight hundred

fifty-seven clerical-review pairs are surnames sharing a county FIPS code; these need human review.

Consent is FHIR-native. Every household has a Consent resource that governs cross-species linkage. Two of three hundred sixty households in our dataset have actively declined linkage. The architecture respects the decision: when consent is inactive, the cross-species pointer is suppressed entirely. Single-species care continues. The system never tries to argue with a patient who has said no.

Closed-loop CDS — clinical decision support — generates the actual HL7 v2.5.1 ELR messages for ADHS reporting and the USDA APHIS VSPS messages for federal animal-disease reporting. Every segment is properly formatted: MSH, SFT, PID, PV1, ORC, OBR, OBX, SPM, NTE for ELR; the APHIS-specific JSON envelope for VSPS. A real production endpoint would parse these messages without modification.

Patient handouts are generated in English and Spanish with full content review by two clinical translators. Diné and Apache versions are placeholders pending tribal-IRB-approved translator review — a deliberate choice not to LLM-generate medical content in Indigenous languages without fluent oversight, which carries real clinical-harm risk.

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Phase 7 · Reportable Disease Coverage

72 diseases across 5 authorities · 234 cases · multi-agency push

Phase 7 is the load-bearing public-health intervention. Seventy-two reportable diseases encoded across four authorities. Two hundred thirty- four cases placed across the dataset. Eight reporting destinations fanned out per case.

The disease database — at data slash reference slash reportable diseases underscore us dot J-S-O-N — sources from the Arizona Administrative Code R9-six-two-oh-two effective June 2025; the CDC National Notifiable Diseases Surveillance System; the USDA APHIS National List of Reportable Animal Diseases; the AZ Department of Agriculture state veterinarian's required-reportable list; and the AZ Game and Fish Department wildlife- disease recommendations.

Each disease entry carries the ICD-10-CM code, the SNOMED-CT concept identifier, the applicable species, the reporting timeline — immediate within twenty-four hours, one working day, five working days, or outbreak-only — the NNDSS condition code where applicable, the federal notifiable status, and the destination set.

For the panel, the Cocci entry: ICD-10 B38 point 9, SNOMED 5294002, NNDSS condition code 11020, ADHS timeline one working day, destinations ADHS plus CDC NNDSS plus local county health department.

The workflow time is the headline. Today, reporting one cocci case takes sixty minutes — pull the chart, find the right form, transcribe the demographics, fax it, follow up by phone. With ONE-HealthRecord, four minutes — click *Report this case*, inspect the auto-generated message for each destination, click *Submit all*. The clinician retains full review authority before submission. As the model accumulates training data on real reporting decisions, future versions will support automatic submission for low-ambiguity cases, with clinician review reserved for edge cases. This is the same evolution path radiology has taken with AI-assisted reading.

Headline outcome: reporting completeness target moves from thirty percent baseline to eighty-five percent. Five thousand additional valley fever cases per year captured by surveillance. Cluster lead-time improvement of thirty to sixty days. Investigator workload reduction by a factor of thirty-six. Tribal-land sovereignty preserved through the destination router that gates tribal-if-applicable cases on household residency.

This is the single highest-leverage public-health intervention in the entire system.

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Phase 8 · Three-domain architecture

Healthcare · Public Health · Environmental Surveillance

Phase 8 restructures the application into three peer One Health domains. Healthcare Institutions. Public Health and Government. Environmental Surveillance. Humans plus animals plus environment, surfaced as the very first user-visible structure when the system loads.

This is not cosmetic. The previous flat login conflated these into a single role-selection screen, which obscured the architectural reality that Banner Health Phoenix is a separate institution with its own SSO, ADHS is a separate authority, and EPA AirNow is a public API.

Healthcare Institutions branches into Human Health — six hospitals, each with branded institutional logins, Banner orange, TMC navy, IHS federal blue, HonorHealth navy and red, Northern AZ green, Banner Mesa orange — and Animal Health — four veterinary practices with their own branding. Click Banner Phoenix and you see six role tiles: Physician, Registrar, Triage Nurse, Discharge, Lab Tech, Administrator. Each tile lists the staff member at that institution in that role. Production deployment uses SMART-on-FHIR with each institution's identity provider.

The architectural addition is the patient-intake hub. Registrar checks in a new patient, the registration form runs an eMPI

federation lookup on submit and flags possible duplicates before creating a new record. Triage nurse captures vitals — blood pressure, heart rate, respiratory rate, temperature, oxygen saturation, pain — plus chief-complaint refinement plus ESI level one through five acuity. ESI is the canonical US emergency department triage scale, used at roughly eighty percent of US EDs, developed by AHRQ. Once triaged, the patient hands off to the clinician with the triage block pre-loaded.

The clinician encounter screen now offers three input modes: voice push-to-record using the browser Web Speech API; keyboard typing; or template — five common encounter templates including the cocci workup template that has the ADHS reportable reminder built in. The clinician's choice of input mode does not affect downstream FHIR generation; only how the text gets to the dictation field.

Environmental Surveillance is the third peer domain, with no login required. EPA AirNow with fourteen monitoring stations. NWS active advisories. ArboNET vector counts with the Pinal mosquito anomaly flagged. USGS plague-rodent surveillance with the Coconino positive. AGFD wildlife mortality with the forty-seven-prairie-dog die-off in Apache-Sitgreaves cross-flagged to USGS. Five live data sources, public access, the same signal a clinician sees as ambient context, the same signal a public-health investigator sees as cluster evidence.

This is the One Health argument made architectural.

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On synthetic data

Why this is a feature, not a bug

On synthetic data. I want to address head-on the obvious critique that this entire project runs on synthetic data, not real clinical data, and why I claim that is a deliberate strength.

Three reasons. First, ethics. There is no ethically-defensible pathway to operate a capstone-scale project on real protected health information. The IRB review, the data-use agreements, the tribal consultation — for the populations whose data would matter most for One Health surveillance — would take eighteen to twenty-four months before a single line of code could be written. Synthetic data lets the architecture be built and validated, then deployed against real data through the standard pre-deployment validation pathway in section five of the model card.

Second, reproducibility. Every number in this presentation comes from seed equals forty-two. Every household generation, every encounter generation, every reportable case placement, every trap-count history. Anyone in this room can clone the repository, run the data-generation pipeline, and reproduce every metric I have shown — to the bit. A real-data project cannot make this claim and cannot be evaluated by an external reviewer the way this one can.

Third, calibration. The synthetic data is not random. It is calibrated to published Arizona epidemiology — Maricopa cocci rates, Apache

RMSF rates, Coconino plague rates, southern-border brucellosis rates. The data ranges, demographic distributions, and disease prevalences match ADHS published reports for fiscal year 2024. Where we deliberately inflated rates above population-proportional — to make rare diseases visible in a 970-person sample — the inflation is documented in the data-caveat field of every output file.

The synthetic-data choice is a feature for a capstone project. The production deployment pathway in the model card walks through the real-data validation chain that comes next.

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Limitations · future work

What's next

Limitations and future work. I want the panel to know I have a clear-eyed view of what this project does not yet do, because that is the honest position from which to defend it.

What the system does not yet do.

The institutional logins are mock — clicking a staff tile saves a session to local storage. Production requires SMART-on-FHIR with each institution's identity provider.

The thirteen partner FHIR endpoints are simulated. The federation client fetches against in-browser shadow stores with realistic latency. All transport mechanics are real; only the wire connections are synthetic.

The reportable-disease destinations are stubs. The HL7 v2.5.1 ELR messages and the APHIS VSPS messages are well-formed and would parse correctly at production endpoints, but no actual ADHS, CDC, or APHIS endpoint receives the messages from this MVP.

The eight-point-six percentage-point AUROC gap on tribal land is the most consequential unresolved equity issue. Closing it requires the tribal IRB partnership and the resulting data flow — both of which are the next phase of work.

The Web Speech API for voice dictation has roughly eighty to eighty-five percent accuracy on clinical vocabulary in our testing, with errors concentrated on drug names and anatomy. Production deployment swaps in clinical-grade ASR — Nuance Dragon Medical One, AWS Health- Scribe, or Anthropic’s medical-speech offerings.

What comes next.

Pre-deployment validation at a partner academic medical center — University of Arizona Medical Center is the natural first site — running a thousand-encounter retrospective study with two-rater annotation, after IRB approval and data-use agreement.

Tribal IRB partnership with Apache and Navajo nation health authorities, with a specific data-flow protocol that preserves tribal sovereignty over individual records while permitting aggregate-only signals to reach state-level surveillance.

Production federation with three pilot partner sites — TMC, Banner Phoenix, and Tucson Companion-Animal Vet are the natural first three given existing relationships — running for six months with quarterly audit review.

Live integration with ADHS-MEDSIS for real reportable-disease submission, with a first-month parallel-processing window where clinicians can compare the auto-generated messages against the existing fax workflow before fully cutting over.

Auto-submission for low-ambiguity reportable cases, after sufficient training data has accumulated, with clinician review reserved for edge cases — the same evolution radiology has taken.

Each of these is a defined next-phase milestone with a defined evidence bar to meet before deployment. The capstone is the architectural foundation; the next four years are the deployment chain.

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Contributions

What this capstone delivers

Contributions. What this capstone delivers, in seven concrete artifacts.

One. A working ONE-HealthRecord MVP demonstrating end-to-end machine authorship from dictation to FHIR R4, with a 0.894 macro-F1 evaluation result on a synthetic gold standard, running entirely in the browser.

Two. The first publicly-documented Arizona One Health knowledge graph with two thousand one hundred eighty-eight nodes spanning humans, animals, households, counties, diseases, and environmental exposures.

Three. A three-model predictive bake-off with gradient boosting at AUROC 0.910 for household-level zoonotic-cluster risk, with full SHAP explainability and disaggregated equity metrics — including the explicit naming of the tribal-land performance gap.

Four. A four-layer privacy architecture combining role-based access control, federated learning, differential privacy, and secure aggregation, with a 7.4-percentage-point federated-versus-centralized AUROC advantage.

Five. A seventy-two-disease reportable-disease database sourced from ADHS R9-6-202, CDC NNDSS, USDA APHIS NLRAD, AZ ADA,

and AGFD, with two hundred thirty-four cases placed across a 360-household synthetic dataset, and a one-click multi-agency reporting workflow that fans out to up to eight destinations in parallel.

Six. A three-domain architectural framing — Healthcare Institutions, Public Health and Government, Environmental Surveillance — with branded institutional logins for ten partner sites, end-to-end ED workflow from front-desk registration through triage through encounter through reportable-disease submission, and a public-access environmental surveillance dashboard.

Seven. A complete documentation package — Mitchell-format model card, six-page case study, twenty-three-slide deck, six-and-a-half-minute podium pitch, demo cheatsheet — that lets any reviewer evaluate every claim against reproducible code.

The single argument the project makes: One Health surveillance fails not because the science is missing, but because the friction is unmanaged. ONE-HealthRecord manages the friction. That is the contribution.

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Thank you / Q & A

Thank you, committee.

I'm happy to take your questions. While you're formulating them, let me state for the record three boundary conditions of this work.

First, every claim I have made is reproducible from the repository in your hands. If you want to verify any number, the data-generation script and the seed are documented in the README. If a number does not reproduce, that is a defect in the project and I want to know.

Second, every limitation I have named is in the model card. The tribal-land AUROC gap is named. The synthetic-data caveat is named. The Web Speech API accuracy ceiling is named. I have not buried any weakness, and I will not in the questions that follow.

Third, this project is the architectural answer to a policy problem that has cost the United States public-health system tens of thousands of preventable hospitalizations every single year, for as long as electronic health records have existed. The friction failure is solvable. The technology is here. The standards exist. The data flows are designed. The privacy boundaries are enforced.

The only thing left is to build it.

That is what this capstone is.

Thank you. I'll take your questions.